TOPIC:

WATER POLLUTION (RIVERS, SEAS, LAKES)

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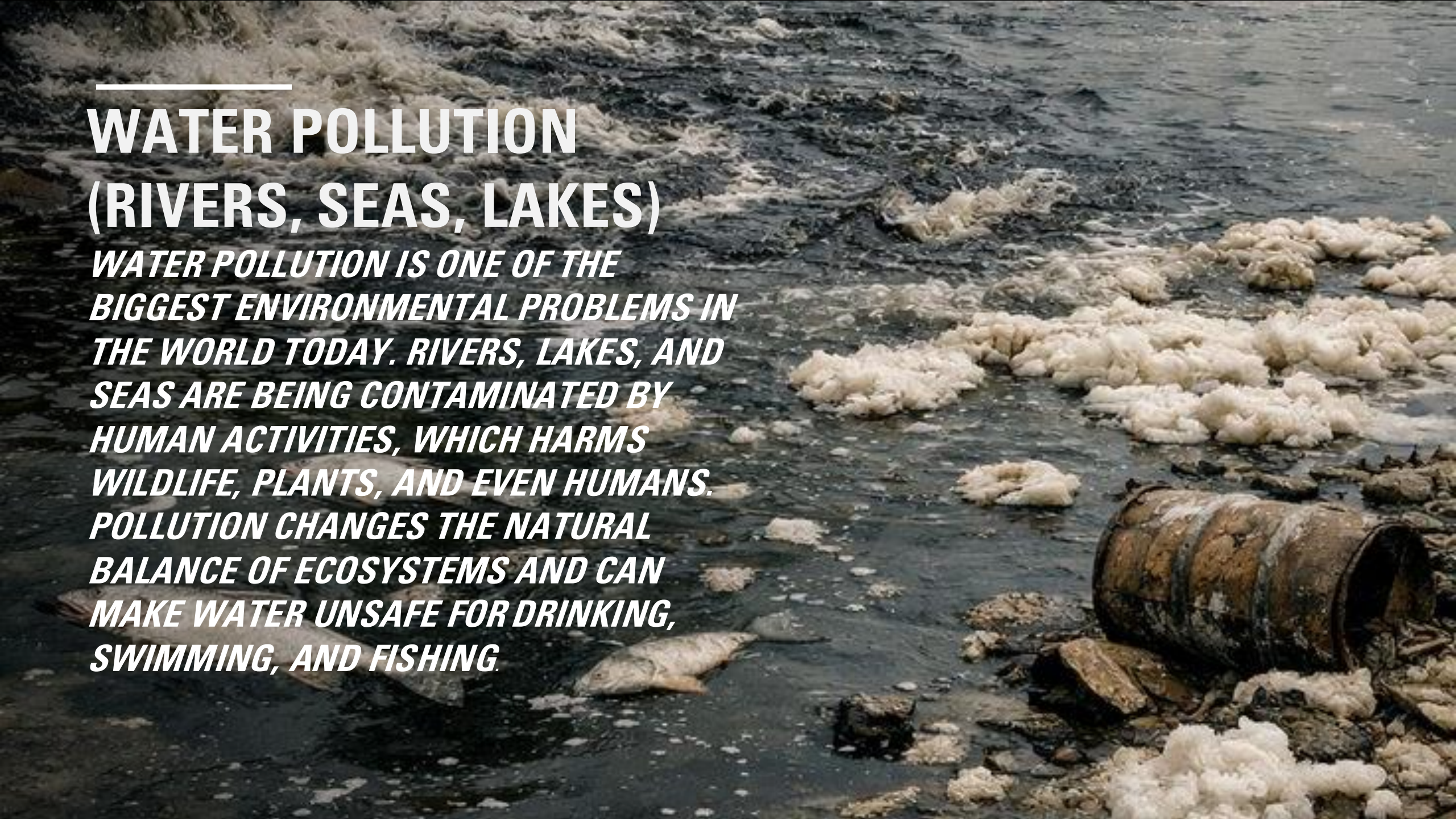
Key Points

What are the main sources of pollution?
How do fertilizers used in agriculture affect the ecosystems of nearby lakes or rivers?
How does plastic enter the sea from land, and how long does it take to fully decompose?
Impact on living organisms (biodiversity)
How do oil spills affect the life cycles of marine birds and mammals?
What threat do microplastics pose to fish and, ultimately, to human health?
Why do certain species of aquatic plants decrease in polluted lakes, and what effect does this have on oxygen levels?



WATER POLLUTION (RIVERS, SEAS, LAKES)

WATER POLLUTION IS ONE OF THE BIGGEST ENVIRONMENTAL PROBLEMS IN THE WORLD TODAY. RIVERS, LAKES, AND SEAS ARE BEING CONTAMINATED BY HUMAN ACTIVITIES, WHICH HARMS WILDLIFE, PLANTS, AND EVEN HUMANS. POLLUTION CHANGES THE NATURAL BALANCE OF ECOSYSTEMS AND CAN MAKE WATER UNSAFE FOR DRINKING, SWIMMING, AND FISHING.

Main Sources of Pollution

The main sources of water pollution include industrial waste from factories, household sewage, and chemicals used in agriculture like fertilizers and pesticides. Plastic waste and oil spills are also major contributors. These pollutants enter water bodies and disrupt natural ecosystems, affecting both plants and animals.

EFFECT OF FERTILIZERS ON NEARBY ECOSYSTEMS

FERTILIZERS USED IN AGRICULTURE OFTEN WASH INTO NEARBY RIVERS AND LAKES. THIS CAUSES EXCESSIVE GROWTH OF ALGAE, KNOWN AS EUTROPHICATION. THE ALGAE USE UP OXYGEN IN THE WATER, WHICH CAN KILL FISH AND OTHER AQUATIC ORGANISMS. OVER TIME, THIS REDUCES BIODIVERSITY AND WEAKENS THE ECOSYSTEM.

HOW PLASTIC REACHES THE SEA

PLASTIC POLLUTION REACHES THE SEA THROUGH RIVERS, WHICH CARRY WASTE FROM LAND, AS WELL AS WIND AND RAIN. HUMAN ACTIVITIES LIKE TOURISM AND SHIPPING ALSO ADD TO THE PROBLEM. PLASTIC TAKES HUNDREDS OF YEARS TO DECOMPOSE, AND SOME PLASTICS NEVER FULLY BREAK DOWN. INSTEAD, THEY TURN INTO MICROPLASTICS, WHICH CONTINUE TO POLLUTE THE WATER.

• Impact on Biodiversity

- Water pollution reduces biodiversity. Many species lose their habitats, and some may become extinct. Food chains are disrupted, and ecosystems become less resilient. Polluted water cannot support the variety of life it once did, which weakens natural processes that are vital for survival.

Effects of Pollution on Marine Life and Ecosystems

WATER POLLUTION HAS SERIOUS IMPACTS ON MARINE LIFE AND ECOSYSTEMS. OIL SPILLS ARE VERY DANGEROUS FOR BIRDS AND MARINE MAMMALS. OIL DAMAGES BIRDS' FEATHERS AND MAMMALS' SKIN, MAKING IT HARD FOR THEM TO KEEP WARM, MOVE, OR FIND FOOD. MANY ANIMALS DIE FROM DIRECT EXPOSURE, WHILE OTHERS SUFFER LONG-TERM HEALTH PROBLEMS.

MICROPLASTICS ARE TINY PIECES OF PLASTIC THAT COME FROM THE BREAKDOWN OF LARGER PLASTICS. FISH OFTEN EAT MICROPLASTICS, WHICH THEN ACCUMULATE IN THEIR BODIES. HUMANS CAN ALSO CONSUME MICROPLASTICS THROUGH SEAFOOD, WHICH MAY HARM HEALTH.

Pollution also affects aquatic plants, which produce oxygen in water. When these plants decline, oxygen levels drop, making survival difficult for fish and other organisms. This chain reaction further damages the ecosystem and reduces overall water quality.

CONCLUSION

WATER POLLUTION IS A SERIOUS GLOBAL PROBLEM THAT REQUIRES IMMEDIATE ACTION. PROTECTING RIVERS, LAKES, AND SEAS IS ESSENTIAL FOR THE HEALTH OF ECOSYSTEMS, ANIMALS, AND HUMANS. EVERYONE HAS A ROLE TO PLAY IN REDUCING POLLUTION AND PRESERVING CLEAN WATER FOR FUTURE GENERATIONS.